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Nadroparin for the prevention of venous thromboembolism in nonsurgical patients: a systematic review and meta-analysis

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Abstract Anticoagulant thromboprophylaxis with low molecular weight heparin is widely used in nonsurgical settings. To obtain best estimates of the effects of nadroparin for the prevention of venous thromboembolism (VTE) in nonsurgical patients, we conducted a systematic review and meta-analysis. Data sources were Medline, Embase, and Cochrane Library supplemented with conference abstracts, without language restrictions. Selection criteria were randomized controlled trials with nadroparin at prophylactic dose in adult nonsurgical patients. Main efficacy outcomes were major VTE (the composite of symptomatic deep vein thrombosis, symptomatic pulmonary embolism, asymptomatic proximal deep vein thrombosis and VTE-related death) and symptomatic VTE. The main safety outcome was major bleeding. We expressed treatment effects as risk ratios. Ten studies (4 vs. placebo or no treatment, 4 vs. UFH, 1 vs. fondaparinux and 1 vs. warfarin) enrolling a total of 7658 patients were included. In comparison with placebo, nadroparin reduced major VTE by about one-half (RR 0.48, 95% CI 0.24–0.97) with a consistent effect on symptomatic VTE (RR 0.69, 95% CI 0.46–1.05) and no increase in major bleeding (RR 1.51, 95% CI 0.40–5.79). In comparison with other pharmacological prophylaxis, nadroparin was similarly efficacious for prevention of major VTE (RR 1.14, 95% CI 0.63–2.10) and symptomatic VTE (RR 1.10, 95% CI 0.51–2.35) and produced similar effects on major bleeding (RR 0.60, 95% CI 0.25–1.50). Five studies were open label, and for three of these the adjudication method was not described or not blinded. In nonsurgical populations at risk of VTE, nadroparin reduced VTE by about one half compared with placebo or no treatment and appeared similarly effective and safe as other prophylactic anticoagulants.

Keywords Low molecular weight heparin · Nadroparin · VTE prevention · Major bleeding · Medical patients · Nonsurgical patients

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Inroduction

Venous thromboembolism (VTE) is a prevalent and avoidable complication in nonsurgical patients with predisposing risk factors. International guidelines recommend anticoagulant prophylaxis with LMWH in high risk acutely ill medical patients and in other nonsurgical populations [1].

Nadroparin calcium was the first LMWH approved for use and is the second most widely used LMWH worldwide. However, the available data with nadroparin for VTE prevention in nonsurgical patients are limited. We pooled the data from randomized controlled trials to obtain best estimates of the effects of nadroparin compared with placebo or active anticoagulant control in nonsurgical populations with respect to clinically important outcomes.

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Methods

Data sources

We attempted to identify all randomized, controlled trials, irrespective of language, that described anticoagulant prophylaxis with nadroparin in nonsurgical patients.

We searched the literature to identify relevant published and unpublished trials, including trials only presented as abstracts. We searched Medline, Embase and the The Cochrane Controlled Trials Register through to July 2015 and the American Society of Hematology and International Society of Thrombosis and Haemostasis congress proceedings using the ISI Web of Knowledge database.

To identify any additional studies, we searched the following databases: Web of science, Core Collection; Journals@Ovid; YOUR Journals@Ovid; Inspec Archive—Science Abstracts; International Pharmaceutical abstracts (EBSCO); and ScienceDirect.

We also searched the internet, including Medscape, the US Food and Drug Administration (FDA) Web site, and the Web sites maintained by the drug manufacturers.

We hand searched reference lists of potentially eligible studies identified from our search of Medline.

Study selection

Two reviewers independently assessed eligibility of studies based on pre-specified inclusion and exclusion criteria. Questions or disagreements were to be resolved by discussion and consensus.

We included a study if it was a randomized, controlled trial that compared nadroparin at prophylactic dose with a comparator (active treatment or placebo/no treatment) in adult patients not undergoing surgery and if it reported pulmonary embolism (PE) and/or deep-vein thrombosis (DVT) of the lower extremities. Trials for the prevention of clotting during hemodialysis or for the treatment of VTE or for the prevention of pregnancy-related complications (e.g., miscarriage) unrelated to VTE were excluded.

The dose of nadroparin was considered to be prophylactic if administered at a total daily dose of 5700 AXa IU or less. Trials at higher doses were excluded. We included studies regardless of treatment duration.

Study data extraction

This systematic review and meta-analysis followed the methods recommended by Cochrane and the PRISMA guidelines [2].

Data regarding inclusion criteria and treatment type and duration were abstracted, when available, from each individual study. We extracted data on the following treatment efficacy outcomes during the treatment period: any PE, fatal PE or VTE-related death, symptomatic DVT, asymptomatic proximal DVT, death due to any cause and major bleeding. We accepted the reported definitions of major bleeding and did not attempt to reclassify these events.

For binary outcomes, numerators were defined as the reported event count and denominators were based on the intent-to-treat (ITT) population (as reported) or on the as-treated population for major bleeding if different.

Study quality assessment

We conducted a risk of bias assessment using the criteria recommended by The Cochrane Collaboration [3]. Two reviewers carried out "risk of bias" assessments, with consensus.

To assess quality, we evaluated studies for potential bias related to random sequence generation, the adequacy of allocation concealment, blinding and blind assessment, incomplete outcome reporting, performance of the primary efficacy analysis according to the ITT principle, as well as other potential sources of potential bias.

Individual criteria were scored as "high risk" if absent, "low risk" if present, or "unclear risk" if absence or presence was ambiguous and reported these assessments in a "risk of bias" table.

We defined a study with a low risk of bias as fulfilling all six of the above criteria and having no major (fatal) flaws. We judged the likelihood of fatal flaws following the Cochrane Handbook for Systematic Reviews of Interventions recommendations [3], including a dropout rate greater than 50% at the follow-up measurement.

Data synthesis and analyses

All statistical analyses were carried out separately for studies with an active comparator and for studies with placebo or no treatment as control.

The treatment effect was measured by the risk ratio (RR). We pooled the RRs on the log scale through the generic inverse variance approach, using both the fixed-effect and the random-effects models.

In the case of a trial reporting no events in both groups, the trial was not taken into account for that outcome since neither the relative risk nor the corresponding variance could be calculated.

To the furthest extent possible and depending on how the data were reported, we carried out analyses on an ITT basis, attempting to include all participants randomized to each group in the analyses, except for major bleeding which was analyzed on an as-treated basis.

Table 1 Main study characteristics

Trial	Studied treatment	Control treatment	Patients	Planned treatment duration	Systematic search of VTE	Sample size nadroparin/ comparator
FONDACAST, 2013	Nadroparin 2850 anti-factor Xa IU	Fondaparinux 2.5 mg (1.5 mg in patients with a creatinine clearance between 30 and 50 ml min^{-1})	Patients requiring rigid or semi-rigid immobilization for isolated non-surgical below-knee injury	At least 21 days, up to 45 days	Compression ultrasound and clinical examination	584/586 (ITT) 622/621 (safety)
PROTECHT (Agnelli), 2009	Subcutaneous injections of nadroparin (3800 IU anti-Xa once a day) up to 4 months	Placebo	Ambulatory patients receiving chemotherapy for metastatic or locally advanced solid cancer	Up to 4 months	Clinical examination	779/387
Mahe et al., 2005	0.3 ml nadroparin (7500 anti-Xa Choay units [2850 IU]) for up to 21 days	Placebo	Bedridden medical in-patients	Up to 21 days	Clinical examination	1230/1244
CIP (Mismetti), 2003	Nadroparin was injected subcutaneously at a fixed daily dose of 2850 IU for 90 days	Warfarin was given orally at a fixed daily dose of 1 mg	Patients with cancer receiving an indwelling long-term central venous catheter	3 months	Clinical examination (venography of upper extremities)	29/30
Fraisse, 2000	Subcutaneous nadroparin adjusted for body weight (0.4 ml, i.e., 3800 AXa IU, or 0.6 ml, i.e., 5700 AXa IU) up to 21 (+1) days	Placebo	Patients mechanically ventilated for acute, decompensated chronic obstructive pulmonary disease	Up to 21 days	Venography	109/114
HSIM (Harenberg), 1996	1 daily subcutaneous administration of LMW heparin (calcium heparin) plus 2 placebo injections for 10 (8–11) days	3 × 5000 IU unfractionated (UF) heparin (calcium heparin) for 10 (8–11) days	Medical inpatients	10 days (8–11 days)	Clinical examination; compression sonography within 24 h of randomization as well as between days 8 and 11 at the end of the study	810/780
Forette et al., 1995	Subcutaneous calcium nadroparin 0.3 ml (3075 anti-Xa IU [2850 IU]) daily for 28 days	Subcutaneous calcium heparin twice a day (0.2 ml-5000 UI—if bodyweight was less than 70 kg or 0.3 ml-7500 IU——if bodyweight was over or equal to 70 kg)	Elderly patients hospitalized with transient locomotor disability	28 days	Clinical examination	146/149
Kujath et al., 1993	Subcutaneous injection of fraxiparin 0.3 ml daily	No thromboprophylaxis	Outpatients with injury of the lower limb immobilized by a plaster cast	During plaster cast period	Compression US	126/127
Aquino et al., 1990	Nadroparin 2850 IU anti Xa 10 days	UFH 10.000 or 15,000 IU (depending on bodyweight) 10 days	Elderly patients with age greater than 70 years	10 days	Doppler ultrasound (confirmed by phlebography)	49/50
Manciet et al., 1990	Nadroparin 2850 IU anti Xa 3 months	UFH (heparin calcium) 5000 IU ×2 or 3 (depending on bodyweight) 3 months	Elderly population with risk factors for DVT	3 months	Doppler ultrasound + phlethysmography confirmed by phlebography	129/127

Statistical heterogeneity was quantified by using the I^2 statistic, which approximates the percentage of the total variation (within and between studies) that is due to between-study variation. A value of 0% indicates no observed heterogeneity whereas larger values show increasing heterogeneity [4].

Results

Study identification and selection

We identified 223 potentially relevant records from the literature search through MEDLINE, EMBASE, and manual bibliography searches after exclusion of duplicate citations. We excluded 189 abstracts during the initial screening, using the predefined inclusion criteria. The remaining 34 records were retrieved for more detailed review and screening. Of these, ten trials met the eligibility criteria and were included in the review and meta-analysis [5–14]. Table 1 shows the main characteristics of the included studies. Reasons for exclusion are summarized in Fig. 1. In one included study [10], VTE outcomes were reported but could not be categorized according to pre-specified definitions and no major bleeding was reported in either arm. As a result, this study did not contribute to the statistical analyses for any outcome.

Study quality

Quality assessment is summarized in Table 2. All trials were randomized, but the method for generating the allocation sequence was adequately reported in only two of ten trials. Seven trials reported concealed treatment allocation. Five studies were double blind. Among the five other studies, two reported an adequate blind outcome assessment. All trials except two provided a description of patient withdrawals.

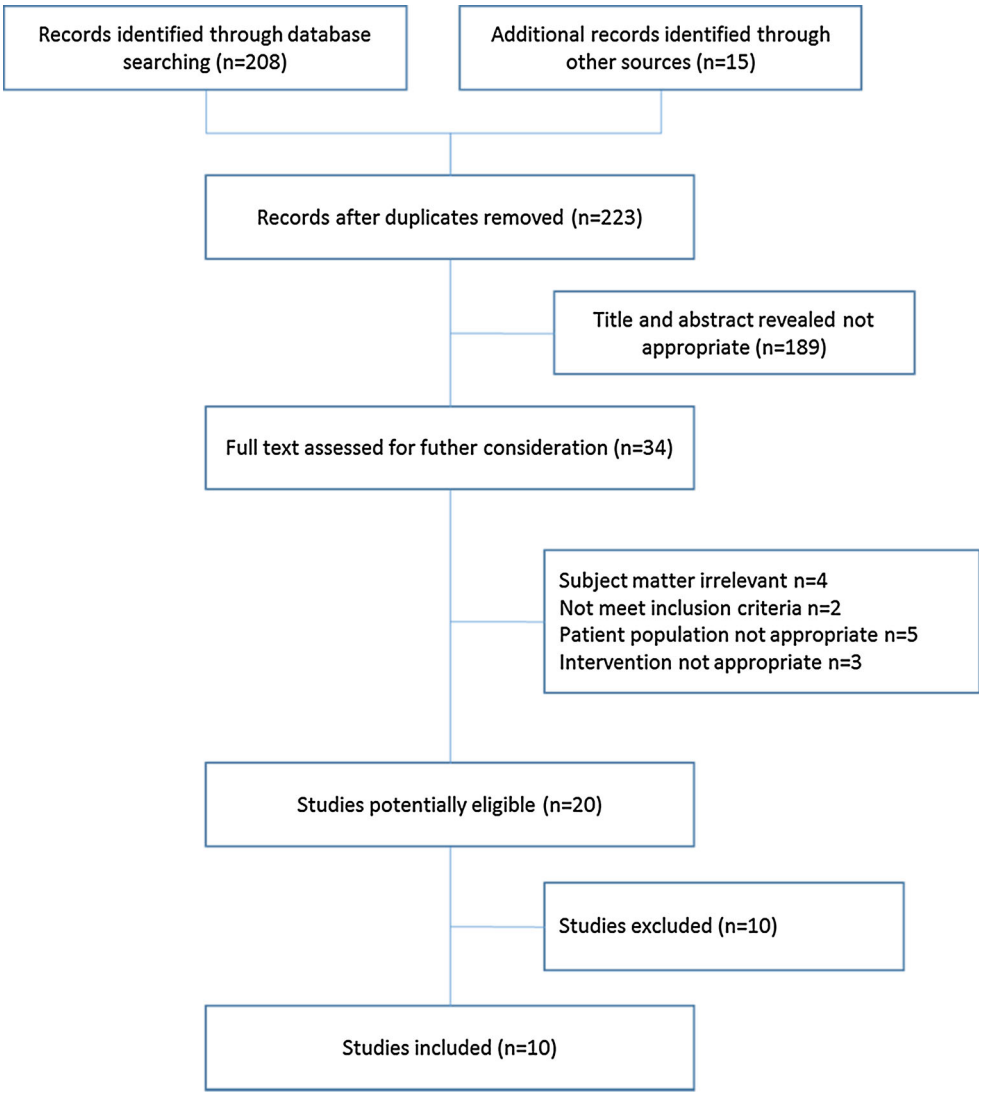

Fig. 1 Search and identification of eligible studies

Outcomes

Nadroparin in comparison with placebo/no treatment

Compared with placebo or no treatment, the use of the nadroparin was associated with a significant reduction in major VTE (composed of symptomatic DVT or PE, asymptomatic proximal DVT and VTE-related death) during the treatment period (RR 0.48, 95% CI 0.24–0.97, $I^2 = 0\%$) (Fig. 2). The effect of nadroparin on symptomatic VTE including VTE-related death was consistent with its effect on major VTE although the difference was not statistically significant (RR 0.69, 95% CI 0.46–1.05, $I^2 = 0\%$) (Fig. 3).

In comparison with placebo, nadroparin use was associated with a higher, although not statistically significant, rate of major bleeding, (RR = 1.51, 95% CI 0.40–5.79). No heterogeneity was found ($I^2 = 24.8\%$) (Fig. 4).

Nadroparin versus other anticoagulants

Control anticoagulants included UFH in 4 studies, fondaparinux in 1 study and warfarin in 1 study. There was no difference in the risk of major VTE between nadroparin and active control treatments (RR 1.14, 95% CI 0.63–2.07, $I^2 = 0$) (Fig. 2). The number of symptomatic VTE including VTE-related death was similar between nadroparin and active control treatments (RR 1.10, 95% CI 0.51–2.35, $I^2 = 0$) (Fig. 3).

A similar rate of major bleeding was observed with nadroparin and with active control (RR 0.60, 95% CI 0.24–1.50) (Fig. 4).

Other outcomes

Additional analyses were carried out on other outcomes during the treatment period: PE (fatal and non-fatal), fatal PE, symptomatic DVT, death due to any cause, the composite of symptomatic VTE and all-cause mortality, non-VTE related death, asymptomatic proximal DVT, and net clinical benefit. Results are summarized in Table 3 for studies comparing nadroparin with placebo or no treatment and in Table 4 for comparisons with active treatment.

Subgroup analyses

Tests for interaction were non-significant in all subgroup analyses, except for the decreased risk of major bleeding with nadroparin in active comparator studies enrolling only elderly patients versus other studies ($p = 0.0235$).

Table 2 Quality assessment

Trial	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of patients	Blinding of the providers	Blinding of the data collectors	Blinding of the outcome adjudicators	Incomplete data outcome reported	Free of selective reporting	Free of other bias
FONDACAST, 2013	Unclear	Yes	No	No	No	Yes	Yes	Yes	Yes
PROTECHT (Agnelli), 2009	Unclear	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mahe et al., 2005	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
CIP (Mismetti), 2003	Yes	Yes	No	No	No	Yes	Yes	Yes	Yes
Fraisse, 2000	Unclear	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
HSIM (Harenberg), 1996	Unclear	Yes	Yes	Yes	Yes	Yes	Yes	No	Unclear
Forette et al., 1995	Unclear	Unclear	No	No	No	No	No	No	Unclear
Kujath et al., 1993	Unclear	Unclear	No	No	No	No	Yes	Yes	Yes
Aquino et al., 1990	Unclear	Unclear	No	No	No	No	No	No	Unclear
Manciet et al., 1990	Unclear	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

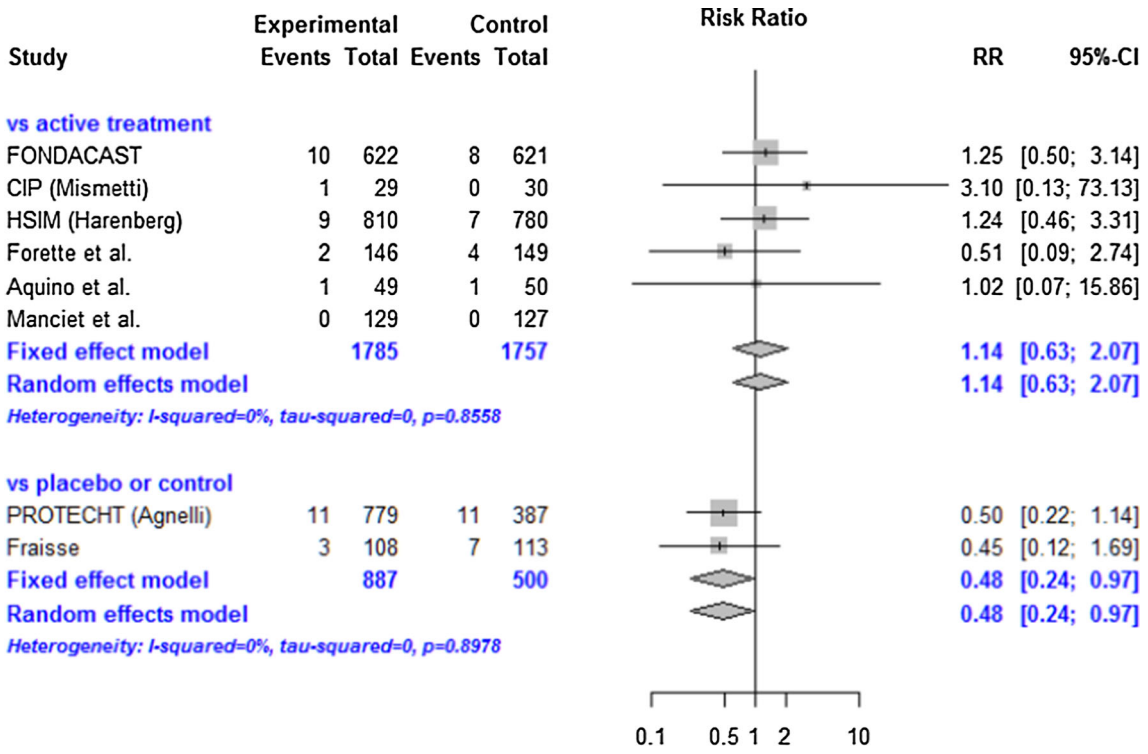

Fig. 2 Major VTE (asymptomatic proximal DVT and symptomatic VTE including fatal PE) during anticoagulant prophylaxis

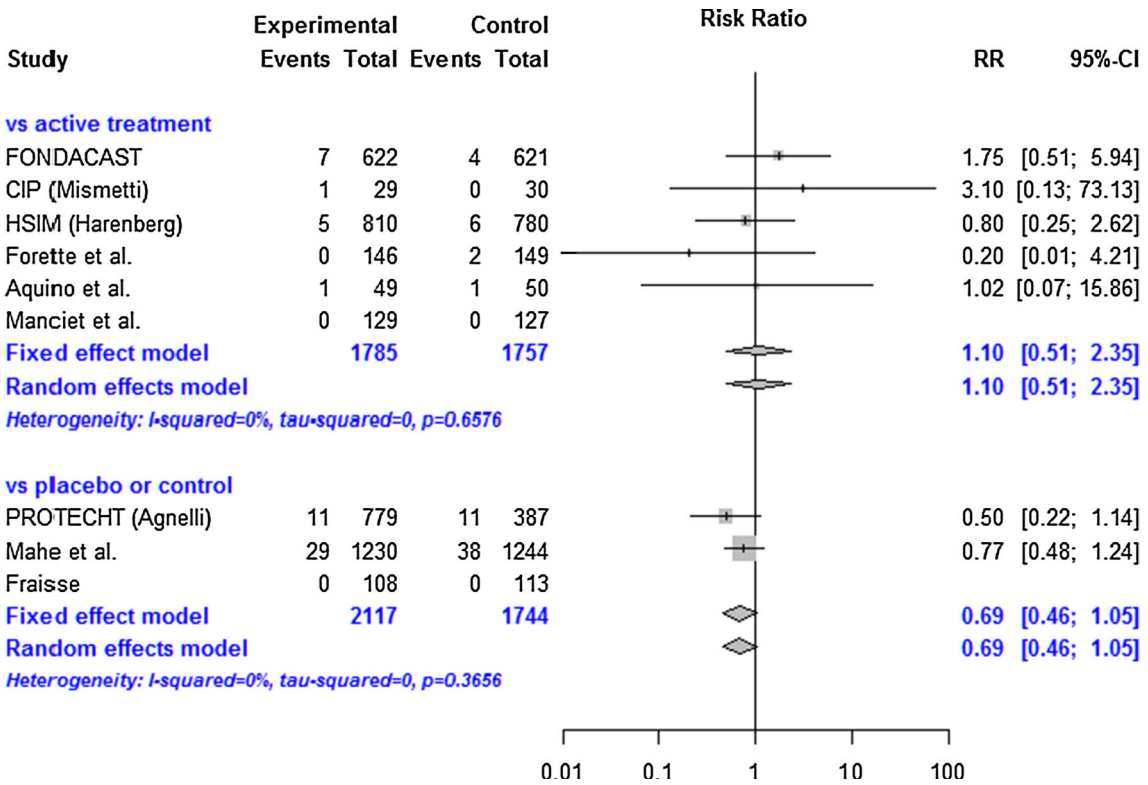

Fig. 3 Symptomatic VTE including Fatal PE during anticoagulant prophylaxis

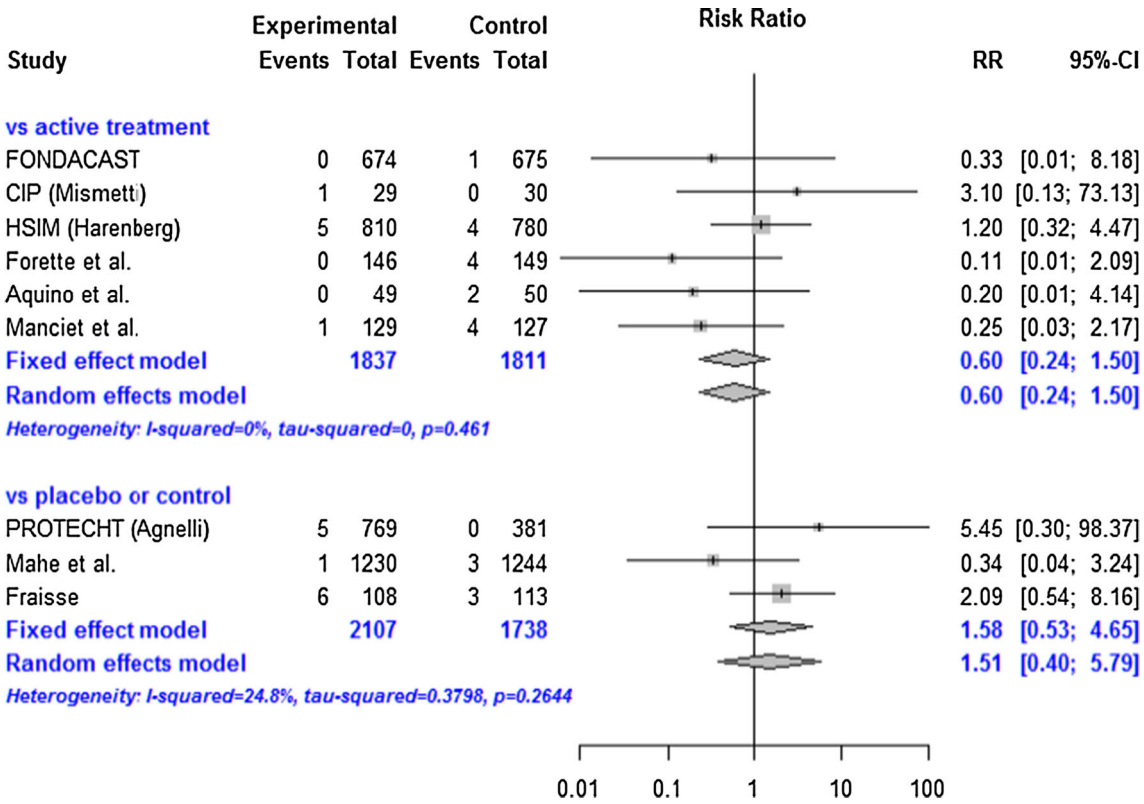

Fig. 4 Major bleeding during anticoagulant prophylaxis

Table 3 Summary of all outcomes (nadroparin versus placebo)

	FE, RR (95% CI)	RE, RR (95% CI)	k	I^2 (%)	Trials
Symptomatic VTE including VTE-related death	0.69 [0.46–1.05]	0.69 [0.46–1.05]	2	0	PROTECHT (Agnelli), Mahe et al., Fraisse
Major VTE (sympt. VTE + asympt. prox. DVT, VTE related deaths)	0.48 [0.24–0.97]	0.48 [0.24–0.97]	2	0	PROTECHT (Agnelli), Fraisse
Symptomatic DVT	0.77 [0.46–1.29]	0.76 [0.44–1.32]	2	8	PROTECHT (Agnelli), Mahe et al., Fraisse
PE (fatal and non fatal)	0.57 [0.29–1.16]	0.57 [0.29–1.16]	2	0	PROTECHT (Agnelli), Mahe et al., Fraisse
Fatal PE only	0.59 [0.27–1.29]	0.59 [0.27–1.29]	1	NA	PROTECHT (Agnelli), Mahe et al., Fraisse
Symptomatic VTE and all cause mortality	0.94 [0.78–1.14]	0.94 [0.78–1.14]	3	0	PROTECHT (Agnelli), Mahe et al., Fraisse
All-cause death	0.99 [0.80–1.22]	0.99 [0.80–1.22]	3	0	PROTECHT (Agnelli), Mahe et al., Fraisse
VTE-related death	NA [NA; NA]	NA [NA; NA]	0	NA	None
Non-VTE death	NA [NA; NA]	NA [NA; NA]	0	NA	None
Asymptomatic proximal DVT	0.43 [0.12–1.62]	0.43 [0.12–1.62]	1	NA	Fraisse
Major bleeding	1.58 [0.53–4.65]	1.51 [0.40–5.79]	3	25	PROTECHT (Agnelli), Mahe et al., Fraisse, Kujath et al.
Net clinical benefit	0.97 [0.80–1.17]	0.97 [0.80–1.17]	3	0	PROTECHT (Agnelli), Mahe et al., Fraisse

Heterogeneity

No heterogeneity was found for any endpoint in comparisons in placebo or no treatment-controlled studies or active comparator studies.

Discussion

The principal findings of this study are that nadroparin is effective for prevention of major VTE and appears to be similarly effective and safe as other prophylactic anticoagulants.

Table 4 Summary of all outcomes (nadroparin versus active comparator)

	FE, RR (95% CI)	RE, RR (95% CI)	k	I^2 (%)	Trials
Symptomatic VTE including VTE-related death	1.10 [0.51–2.3]	1.10 [0.512–2.3]	5	0	FONDACAST, CIP (Mismetti), HSIM (Harenberg), Forette et al., Aquino et al., Manciet et al.
Major VTE (sympt. VTE + asympt. prox. DVT, VTE related deaths)	1.14 [0.63–2.1]	1.14 [0.625–2.1]	5	0	FONDACAST, CIP (Mismetti), HSIM (Harenberg), Forette et al., Aquino et al., Manciet et al.
Symptomatic DVT	2.02 [0.63–6.5]	2.02 [0.631–6.5]	4	0	FONDACAST, CIP (Mismetti), HSIM (Harenberg), Forette et al., Aquino et al., Manciet et al.
PE (fatal and non fatal)	0.62 [0.22–1.8]	0.62 [0.219–1.8]	3	0	FONDACAST, CIP (Mismetti), HSIM (Harenberg), Forette et al., Aquino et al., Manciet et al.
Fatal PE only	1.28 [0.29–5.7]	1.28 [0.288–5.7]	1	NA	FONDACAST, CIP (Mismetti), HSIM (Harenberg), Forette et al., Aquino et al., Manciet et al.
Symptomatic VTE and all cause mortality	1.29 [0.86–1.9]	1.29 [0.855–1.9]	6	0	FONDACAST, CIP (Mismetti), HSIM (Harenberg), Forette et al., Aquino et al., Manciet et al.
All-cause death	1.37 [0.86–2.2]	1.31 [0.784–2.2]	6	13	FONDACAST, CIP (Mismetti), HSIM (Harenberg), Forette et al., Aquino et al., Manciet et al.
VTE-related death	1.28 [0.29–5.7]	1.28 [0.288–5.7]	1	NA	FONDACAST, HSIM (Harenberg), Aquino et al., Manciet et al.
Non-VTE death	1.51 [0.80–2.8]	1.21 [0.429–3.4]	4	47	FONDACAST, HSIM (Harenberg), Aquino et al., Manciet et al.
Asymptomatic proximal DVT	1.19 [0.42–3.4]	1.19 [0.42–3.4]	3	0	FONDACAST, HSIM (Harenberg), Forette et al., Manciet et al.
Major bleeding	0.60 [0.24–1.5]	0.60 [0.244–1.5]	6	0	FONDACAST, CIP (Mismetti), HSIM (Harenberg), Forette et al., Aquino et al., Manciet et al.
Net clinical benefit	1.07 [0.74–1.6]	0.96 [0.550–1.7]	6	47	FONDACAST, CIP (Mismetti), HSIM (Harenberg), Forette et al., Aquino et al., Manciet et al.

The magnitude of the risk reduction when comparing nadroparin with placebo or no treatment was similar to that observed in the meta-analysis by Dentali and colleagues comparing anticoagulant prophylaxis (enoxaparin, dalteparin, or fondaparinux) with placebo for prevention of symptomatic DVT in hospitalized medical patients (RR 0.47, CI 0.22–1.00) [15].

The relative risk of major bleeding with nadroparin versus placebo or no treatment was similar to that reported with any LMWH, UFH and fondaparinux compared with placebo in the ACCP consensus (OR 1.32; 95% CI 0.73–2.37) [1]. Rates of major bleeding with nadroparin were similar to active control, which is also consistent with the findings of meta-analyses comparing LMWH with active control [1, 16].

The apparent lack of effect of nadroparin on all-cause mortality is consistent with the findings of the meta-analysis by Dentali et al on anticoagulant prophylaxis with LMWH (including for this outcome enoxaparin, dalteparin, nadroparin, or fondaparinux) in hospitalized medical patients [1, 15] and of a subsequent large trial [17] with enoxaparin.

We included all relevant nonsurgical studies and thus a broader population than hospitalized medical patients alone. While the underlying risk of VTE and the duration of treatment varied among studies, no interaction (data not shown) was detected in subgroup analyses comparing treatment effects in studies enrolling medical patients versus studies enrolling other non-surgical patients.

Five of the ten trials included in this meta-analysis were not blinded. Other potential limitations are the differences among the trials in doses of nadroparin and of comparators, the duration of prophylaxis, use of mandatory DVT screening methods, and definitions of major bleeding. However, no substantial heterogeneity was found for any comparisons or endpoints.

The strengths of this study include a rigorous methodological approach, the selection of all VTE prevention studies conducted with nadroparin at prophylactic dose, the consistency of the results across trials, and the focus on patient important outcomes.

This the first systematic review and meta-analysis of nadroparin for the prevention of VTE in nonsurgical patients. The results indicate that nadroparin is effective and safe for prevention of VTE in nonsurgical patients at risk of VTE.

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Authors Contribution WA, JB, and JWE reviewed the data and wrote the article. MC carried out the statistical analyses and contributed to sections on methods and study quality and identification.

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Compliance with ethical standards

Conflicts of interests Drs. Ageno, Cucherat and Eikelboom have received honoraria and research support from companies that market anticoagulants for prevention of VTE.

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